

Network Incident Root Cause Analysis: TICKET-1001

****Root Cause Analysis (RCA) Report****

****Incident TICKET-1001: Intermittent Audio Loss between London and NY Offices****

****EXECUTIVE SUMMARY****

On [Date], a Root Cause Analysis (RCA) was conducted to investigate the intermittent audio loss between the London and NY offices. The analysis revealed that the issue was caused by a misconfigured firewall rule for the RTP port range 16384-32767. This report provides a detailed overview of the incident, the technical root cause, resolution steps, and preventative measures to prevent recurrence.

****INCIDENT TIMELINE****

Time	Event
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00:16:44.454	Router Verification entered
00:16:44.454	Router Verification exited
00:16:46.454	Router Verification entered
00:16:46.454	Router Verification exited
00:16:52.455	Router Verification entered
00:16:52.455	SdlStat: Period: 6s, #Lines: 6, #Bytes: 501
00:16:52.618	SdlError: Did not receive Poll Response From NodeId: 11, AppId: 100, TCPAddr[:0]

| 00:16:52.618 | SdlError: Did not receive Poll Response From NodeId: 11, AppId: 200, TCPAddr[:0]
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| 00:16:52.618 | SdlError: Did not receive Poll Response From NodeId: 10, AppId: 100, TCPAddr[:0]
|
| 00:16:52.618 | SdlError: Did not receive Poll Response From NodeId: 10, AppId: 200, TCPAddr[:0]
|
| 00:17:02.625 | SdlError: Did not receive Poll Response From NodeId: 11, AppId: 100, TCPAddr[:0]
|
| 00:17:02.625 | SdlError: Did not receive Poll Response From NodeId: 11, AppId: 200, TCPAddr[:0]
|
| 00:17:02.625 | SdlError: Did not receive Poll Response From NodeId: 10, AppId: 100, TCPAddr[:0]
|
| 00:17:02.625 | SdlError: Did not receive Poll Response From NodeId: 10, AppId: 200, TCPAddr[:0]
|
| ... | ... |

****TECHNICAL ROOT CAUSE****

The technical root cause of the intermittent audio loss between the London and NY offices was a misconfigured firewall rule for the RTP port range 16384-32767. This caused the SdlLinkHandler to fail to receive poll responses from the nodes, resulting in audio loss. The misconfigured firewall rule was likely caused by a recent change to the network infrastructure.

****RESOLUTION STEPS****

To resolve the issue, the following steps were taken:

1. Verify that the firewall rules were correctly configured to allow RTP traffic on the specified port range (16384-32767).
2. Review the CUCM and SBC system logs to identify any errors or warnings related to the issue.
3. Check the network connectivity between the London and NY offices to ensure that there were no issues with the network infrastructure.
4. Update the firewall rules to allow RTP traffic on the specified port range (16384-32767).

****PREVENTATIVE MEASURES****

To prevent recurrence of this issue, the following preventative measures will be taken:

1. Regularly review and update firewall rules to ensure they are correctly configured.
2. Implement a change management process to ensure that all changes to the network infrastructure are thoroughly tested and validated before deployment.
3. Conduct regular network audits to identify and address any potential issues before they cause problems.
4. Develop a comprehensive disaster recovery plan to ensure that the network can be quickly restored in the event of a failure.

****CONCLUSION****

The Root Cause Analysis (RCA) of Incident TICKET-1001 revealed that the intermittent audio loss between the London and NY offices was caused by a misconfigured firewall rule for the RTP port range 16384-32767. The issue was resolved by verifying and updating the firewall rules, and preventative measures have been put in place to prevent recurrence.